ISMAIL ALTAY ATAMAN

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SUMMARY

Software Engineer with 3 years of professional experience in IoT, embedded, and full-stack systems. Specializes in C/C++, IoT architecture, AWS IoT Core, Azure IoT Hub, and Flutter. Experienced in multi-node CAN bus protocols, fail-safe OTA pipelines, diagnostic tools, and production-ready applications. Builds reliable, field-ready products using AI-assisted development workflows.

WORK EXPERIENCE

Embedded and IoT Solutions Engineer, Nerva / Lemi Group Inc. (US-Based Stealth Mobility Startup) **(October 2023 – August 2026)**

- Architected scalable telemetry systems using AWS IoT Core and Azure IoT Hub with automated Fleet Provisioning, supporting real-time monitoring across a growing device fleet.
- Developed C++ firmware for cellular IoT devices, implementing fail-safe OTA updates that significantly reduced on-site field maintenance interventions.
- Designed a custom multi-node CAN bus protocol spanning a three-zone hardware architecture for a mobility device, coordinating real-time communication between motor controllers, an IoT gateway, and an SBC-based HMI.
- Built cross-platform HMI and diagnostic tools using Flutter targeting embedded Linux (Rockchip RK3562), visualizing low-level hardware telemetry for a wheelchair control system.

Intern Engineer, Baykar Technologies **(July – August 2023)**

- Developed automated test scripts using Winium to verify UAV ground-station mission-control software, significantly reducing manual regression testing time.

Intern Engineer, DeltaV Space Technologies **(January – February 2023)**

- Designed and tuned discrete PID control algorithms for a 3-axis reaction wheel assembly, stabilizing payload orientation under simulated disturbances.

Intern Engineer, TÜBİTAK SAGE **(August – September 2022)**

- Built GUIs for embedded defense systems using Qt and C++, improving operator interaction efficiency.

PROJECTS

• **yiy.io | Offline-First POS & Automation Platform** (Project Page)

- Architected a zero-cloud-dependency, offline-first POS platform (Node.js/SQLite) running locally to ensure uninterrupted venue operations and zero cloud latency.
- Built a React/TypeScript admin dashboard for real-time order analytics alongside a Flutter waiter app that pairs via local Wi-Fi QR sync.
- Packaged full-stack architecture into a single standalone executable (.exe) for one-click deployment without external database setup.

• **GradeMyJob | AI Job Match & Analytics Extension** (Project Page)

- Architected a cross-browser extension (Chrome MV3 / Firefox MV2) using React 18, TypeScript, and WXT, featuring real-time DOM scrapers across LinkedIn, Indeed, and We Work Remotely.
- Built a multi-tier LLM extraction pipeline integrating Google Gemini and OpenAI with automated model fallbacks and structured JSON parsing to evaluate job fit, red/green flags, and technical requirements.
- Engineered a serverless cloud backend on AWS Lambda and DynamoDB with TTL-based request caching to reduce API latency and minimize LLM token consumption.

• **Yildiz Rocket Team | Team Leader & Avionics Lead**

- Achieved Finalist status at Spaceport America Cup, leading a 50+ person cross-functional team to design and launch high-altitude rockets reaching 20,000 ft.
- Designed the avionics architecture and control software for hybrid rocket motor propulsion, integrating real-time sensor fusion and parachute deployment logic.

- **TÜBİTAK 2209A Project:** Modeled control systems for spacecraft atmospheric landing simulations using MATLAB/Simulink.

SKILLS

- **Languages:** Turkish (Native), English (C1)
- **Programming:** C, C++, C#, Flutter/Dart, Python, JavaScript/TypeScript (Node.js/Automation), MATLAB
- **Embedded Systems & Hardware:** ESP32 (ESP-IDF), STM32, FreeRTOS, Cellular Modems (GSM/LTE/Quectel), GNSS
- **Protocols & Architecture:** CAN Bus 2.0B, UART, SPI, I2C, TCP/IP, MQTT, WebSockets, RESTful APIs, Local Network Sync
- **Cloud & IoT:** AWS IoT Core, Azure IoT Hub, Fleet Provisioning, OTA Update Architecture, SQLite (Edge Storage)
- **Tools & DevOps:** Git, GitHub Actions (CI/CD), CMake, Linux CLI, Altium Designer, VS Code
- **Domain Expertise:** Embedded/IoT Systems, Mobility/Control Systems, Avionics, Full-Stack Local Automation

EDUCATION

Yildiz Technical University (Istanbul, Turkey)

- Mechatronics Engineering - 2025